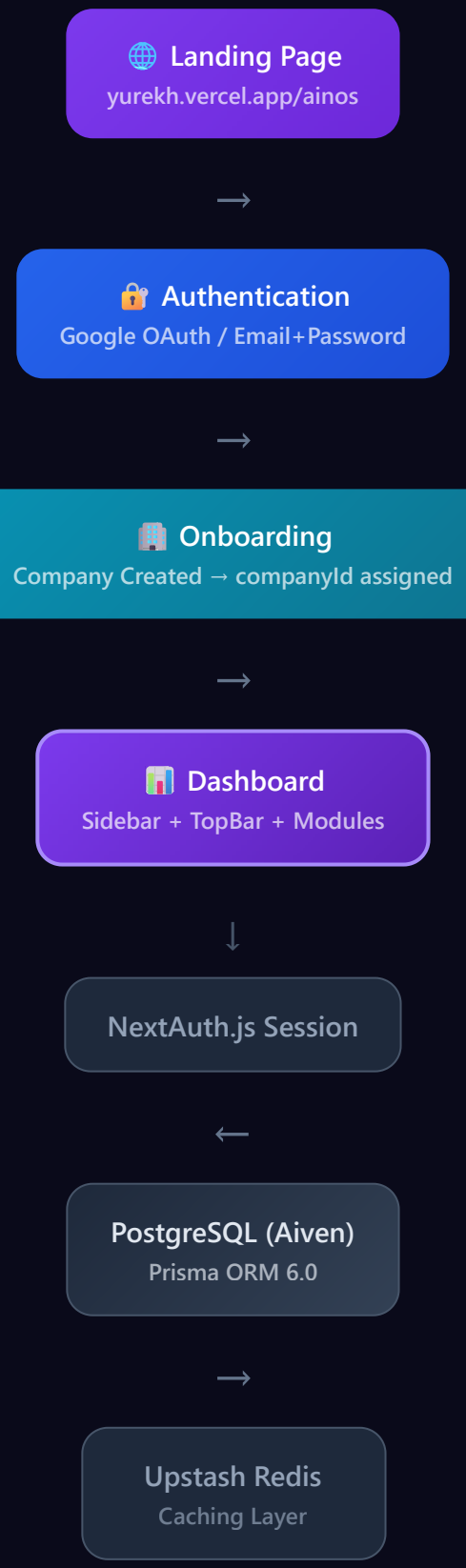
AINOS Platform Architecture

Unified Business Suite — How Everything Connects

1 — USER JOURNEY FLOW



2 — BUSINESS SUITE MODULES (ALL COMPANYID-ISOLATED)

📇 CRM

- Contacts — lead pipeline management
- Leads — scoring, source tracking
- Deals — value, stage, probability
- Follow-Ups — automated reminders
- Customers — GST/PAN profiles

`/api/contacts · /api/leads · /api/deals`

💰 Finance & Billing

- Invoices — GST auto-split, CGST/SGST/IGST
- Quotes — estimate-to-invoice flow
- Sales Orders — order lifecycle
- Expenses — category-wise tracking
- Delivery Challans — GST-compliant
- Tally Import — Excel → Invoice via SheetJS
- Payment Reminders — automated
- Invoice Templates — custom branding

`/api/invoices · /api/quotes · /api/expenses`

👥 HR & Payroll

- Employees — profiles, departments, salary
- Attendance — check-in / check-out
- Leave Requests — approval workflow
- Payroll Runs — period-based calculation
- Timesheets — project-hour logging
- Recruitment — candidate pipeline

`/api/employees · /api/payroll · /api/attendance`

📦 Inventory

- Products — SKU, HSN, barcode, GST rate
- Stock Items — quantity, min/max, reorder
- Vendors — GST, payment terms
- Warehouses — location, capacity
- Purchase Orders — vendor ordering
- Sales Orders — fulfillment tracking

`/api/products · /api/stock · /api/vendors`

📣 Marketing

- AI Blog Agent — auto-generate + schedule
- Blog Posts — CRUD, SEO, analytics
- Email Campaigns — drip sequences
- SEO Tools — keyword research, site audit
- Invitations — design + customize
- Social Captions — AI-generated
- Content Clusters + Internal Links

`/api/blog-posts · /api/blog-agent/* · /api/seo/*`

🤖 AI Studio

- AI Media — Image/Video/Cinema generation
- Video Effects — 18 viral presets
- Lip Sync Studio — AI lip-sync
- AI Chat — context-aware assistant
- AI Skills — specialized tools
- AI Projects — community showcase

`/api/ai-media · /api/ai-chat · muapi.ts`

🎧 Support

- Helpdesk — ticket tracking, priority
- Knowledge Base — articles, categories
- Service Requests — Yurekh Services catalog

- Calendar — events, attendees
- Documents — file management

`/api/helpdesk · /api/knowledge-base`

📋 Projects & Tasks

- Projects — budget, manager, timeline
- Tasks — priority, assignment, due dates
- Automations — trigger → action rules
- Compliance — deadline alerts
- Reports — cross-module insights

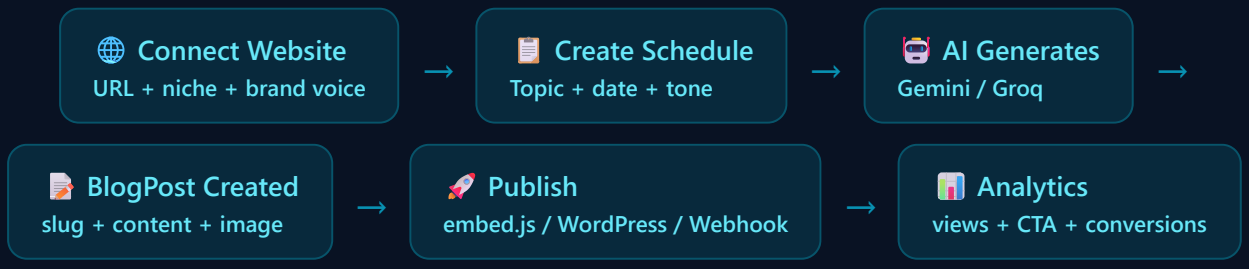
`/api/projects · /api/tasks · /api/automations`

💳 Billing & Credits

- Credit Balance — per-company ledger
- Credit Transactions — audit trail
- Razorpay Checkout — payment webhook
- Subscriptions — plan management
- Tool Runs — credit cost per execution

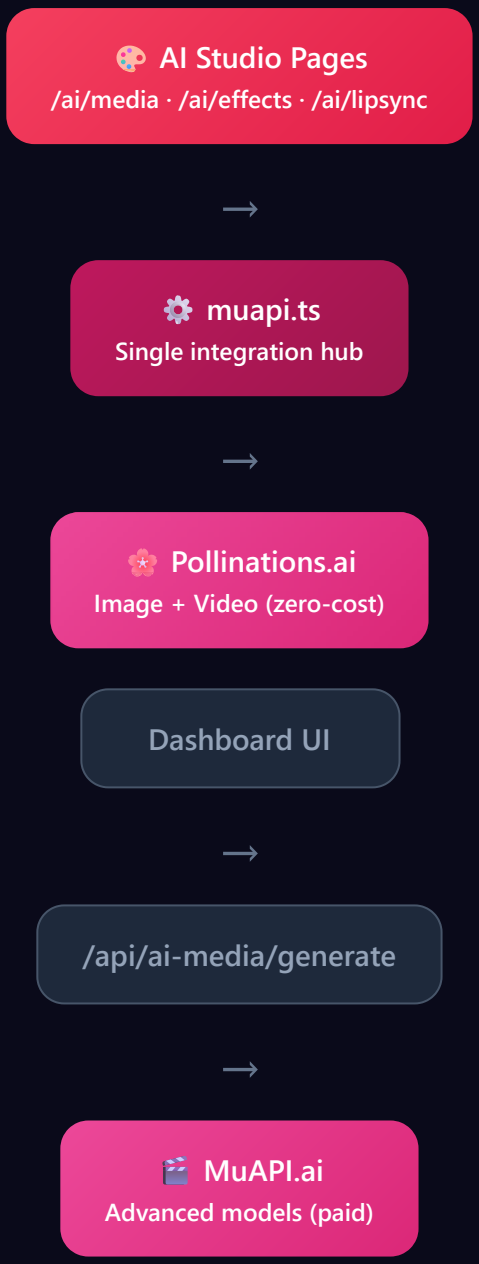
`/api/billing/* · /api/webhooks/razorpay`

3 — AI BLOG AGENT PIPELINE



ConnectedWebsite → BlogSubscription → BlogSchedule → BlogPost

4 — AI GENERATION ARCHITECTURE



5 — MULTI-TENANT ISOLATION MODEL

🔒 companyId Isolation — Every Query, Every Time

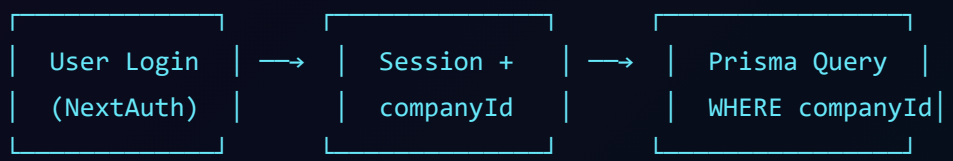
Every business model carries a

`companyId`

field. Every authenticated API route reads the session user's

`companyId`

and filters its Prisma queries by it — so Company A's data never leaks into Company B's dashboard.



6 — TECH STACK

Next.js App Router
Framework — SSR + API Routes

Prisma ORM 6.0
Type-safe database access

PostgreSQL (Aiven)
Primary data store

Upstash Redis
Caching + rate limiting

NextAuth.js
Google OAuth + credentials

Tailwind CSS
Utility-first styling

Pollinations.ai
Zero-cost AI image/video

MuAPI.ai
Advanced AI generation

Razorpay
Payment gateway

Render / Vercel
Deployment platforms

7 — DATABASE MODELS (40+ PRISMA MODELS)

CRM

Contact · Lead · Deal · Customer · FollowUp

FINANCE

HR

Invoice · Quote · Expense · SalesOrder · DeliveryChallan · InvoiceTemplate

Employee · Attendance · LeaveRequest · PayrollRun · Timesheet · Candidate

INVENTORY

Product · StockItem · Vendor · Warehouse · PurchaseOrder

MARKETING

BlogPost · BlogSchedule · BlogSubscription · ConnectedWebsite · EmailCampaign

AI

AIChatSession · ToolRun · Tool · GeneratedSite

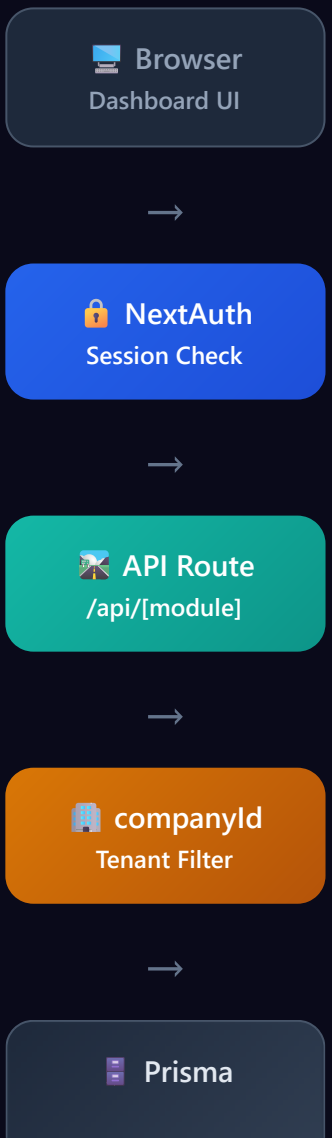
PLATFORM

Company · User · Subscription · CreditBalance · CreditTransaction · Notification

PROJECTS

Project · Task · Automation · ComplianceTask · CalendarEvent · Document

8 — API REQUEST LIFECYCLE



PostgreSQL



📊 Response

JSON → UI